

Early History of Multimedia

A brief history of the use of multimedia to communicate ideas might begin with newspapers, which were perhaps the first mass communication medium, using text, graphics, and images. Before still-image camera was invented, these graphics and images were generally hand-drawn. Joseph Nicéphore Niépce captured the first natural image from his window in 1826 using a sliding wooden box camera. It was made using an 8-h exposure on pewter coated with bitumen. Later, Alphonse Giroux built the first commercial camera with a double-box design. It had an outer box fitted with a landscape lens, and an inner box holding a ground glass focusing screen and image plate. Sliding the inner box makes the objects of different distances be focused. Similar cameras were used for exposing wet silver-surfaced copper plates, commercially introduced in 1839. In the 1870s, wet plates were replaced by the more convenient dry plates. Figure 1.1 (image from author's own collection) shows an example of a nineteenth century dry-plate camera, with bellows for focusing. By the end of the nineteenth century, film-based cameras were introduced, which soon became dominant until replaced by digital cameras.



Fig. 1.1: A vintage dry-plate camera. E&H T Anthony model Champion, circa 1890

Thomas Alva Edison's phonograph, invented in 1877, was the first device that was able to record and reproduce sound. It originally recorded sound onto a tinfoil sheet phonograph cylinder. Figure 1.2 shows an example of an Edison's phonograph (Edison GEM, 1905; image from author's own collection).



Fig. 1.2: An Edison phonograph, model GEM.

The phonographs were later improved by Alexander Graham Bell. Most notable improvements include wax-coated cardboard cylinders, and a cutting stylus that moved from side to side in a “zig zag” pattern across the record. Emile Berliner further transformed the phonograph cylinders to gramophone records. Each side of such a flat disk has a spiral groove running from the periphery to near the center, which can be conveniently played by a turntable with a tonearm and a stylus. These components were improved over time in the twentieth century, which eventually enabled quality sound reproducing that is very close to the origin. The gramophone record was one of the dominant audio recording formats throughout much of the twentieth century. From the mid-1980s, phonograph use declined sharply because of the rise of audio tapes, and later the Compact Disc (CD) and other digital recording formats. Figure 1.3 shows the evolution of audio storage media, starting from the Edison cylinder record, to the flat vinyl record, to magnetic tapes (reel-to-reel and cassette), and modern digital CD.



Fig. 1.3: Evolution of audio storage media. Left to right an Edison cylinder record, a flat vinyl record, a reel-to-reel magnetic tape, a cassette tape, and a CD

Motion pictures were originally conceived of in the 1830s to observe motion too rapid for perception by the human eye. Edison again commissioned the invention of a motion picture camera in 1887. Silent feature films appeared from 1910 to 1927; the silent era effectively ended with the release of *The Jazz Singer* in 1927.

In 1895, Guglielmo Marconi conducted the first wireless radio transmission at Pontecchio, Italy, and a few years later (1901), he detected radio waves beamed across the Atlantic. Initially invented for telegraph, radio is now a major medium for audio broadcasting. In 1909, Marconi shared the Nobel Prize for Physics.

Television, or TV for short, was the new medium for the twentieth century. In 1884, Paul Gottlieb Nipkow, a 23-year-old university student in Germany, patented the first electromechanical television system which employed a spinning disk with a series of holes spiraling toward the center. The holes were spaced at equal angular intervals such that, in a single rotation, the disk would allow light to pass through each hole and onto a light-sensitive selenium sensor which produced the electrical pulses. As an image was focused on the rotating disk, each hole captured a horizontal “slice” of the whole image. Nipkow’s design would not be practical until advances in amplifier tube technology, in particular, the cathode ray tube (CRT), became available in 1907. Commercially available since the late 1920s, CRT-based TV established video as a commonly available medium and has since changed the world of mass communication.

All these media mentioned above are in the analog format, for which the time-varying feature (variable) of the signal is a continuous representation of the input, i.e., analogous to the input

audio, image, or video signal. The connection between computers and digital media, i.e., media data represented using the discrete binary format, emerged actually only over a short period:

- 1967
 - Nicholas Negroponte formed the Architecture Machine Group at MIT.
- 1969
 - Nelson and van Dam at Brown University created an early hypertext editor called FRESS. The present-day Intermedia project by the Institute for Research in Information and Scholarship (IRIS) at Brown is the descendant of that early system.
- 1976
 - The MIT Architecture Machine Group proposed a project entitled “Multiple Media.” This resulted in the Aspen Movie Map, the first videodisk, in 1978.
- 1982
 - The Compact Disc (CD) was made commercially available by Philips and Sony, which was soon becoming the standard and popular medium for digital audio data, replacing the analog magnetic tape.
- 1985
 - Negroponte and Wiesner co-founded the MIT Media Lab, a leading research institution investigating digital video and multimedia.
- 1990
 - Kristina Hooper Woolsey headed the Apple Multimedia Lab, with a staff of 100. Education was a chief goal.
- 1991
 - MPEG-1 was approved as an international standard for digital video. Its further development led to newer standards, MPEG-2, MPEG-4, and further MPEGs, in the 1990s.
- 1991
 - The introduction of PDAs in 1991 began a new period in the use of computers in general and multimedia in particular. This development continued in 1996 with the marketing of the first PDA with no keyboard.
- 1992

- JPEG was accepted as the international standard for digital image compression, which remains widely used today (say, by virtually every digital camera).
- 1992
 - The first audio multicast on the multicast backbone (MBone) was made.
- 1995
 - The JAVA language was created for platform-independent application development, which was widely used for developing multimedia applications.
- 1996
 - DVD video was introduced; high-quality, full-length movies were distributed on a single disk. The DVD format promised to transform the music, gaming, and computer industries.
- 1998
 - Handheld MP3 audio players were introduced to the consumer market, initially with 32 MB of flash memory.